

Polyurethane antifouling coatings with various antifouling strategies in the side chain

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ABSTRACT

Marine fouling is a major barrier to the further development of the marine industry. As science and technology progress, the antifouling coating is still difficult to exhibit feasible and effective strategies against multiple fouling organisms by simple modification. This study reports a polyurethane coating consisting of quaternary ammonium and zwitterions via modifying the imidazole on the side chain, where the former exhibits fouling-degrading performance to kill the bacteria sufficiently, and the latter relies mainly on the formation of the hydrated barrier that could present superior rejection of protein and algae due to the fouling-resistant property. By controlling the number of different antifouling groups in the side chain, the antifouling strategies can be adjusted to deal with marine pollution in different environments. The modified coating could expose hydrophilic and hydrophobic groups on the surface, which produce micro/nano structures underwater. The comprehensive antifouling properties of the coating makes it have a promising future in marine antifouling.

1. Introduction

The ocean as a significant industry for human beings creates \$1.5 trillion in 2010 and is expected to double by 2030 [1], still is a complicated topic about biodiversity, water quality and environmental pollution [1–5]. On the one hand, marine fouling is one of the major losses to the marine economy [5]. On the other hand, the fuel consumption caused by fouling resistance will also have a bad impact on the environment [6]. With the updating of anti-pollution standards, the functional materials that are conducive to environmental protection also gradually are changing to the first choice. Correspondingly, polymer-based marine antifouling coatings have been paid more attention. Nontoxicity, durability, biocompatibility and convenient processability in production are the reasons why these materials will be the first choice in the future [7,8].

The strategies to endow a surface with functional antifouling properties mainly are three ways: preventing foulants from attaching to the surface (fouling-resistant) [9], weakening the interaction between foulants and surface (fouling-release) [10] and degrading/killing biofoulants (fouling-degrading) [7]. Fouling-resistant coatings are mostly based on poly(ethylene glycol) (PEG) and other hydrophilic polymers

(polyacrylates and zwitterions), where highly hydrated surfaces with high interfacial energy prevent the attachment from foulants by coordinating water networks [9]. Moreover, fouling-degrading coatings mainly work on the cell viability and function to achieve the anti-fouling effect, including quaternary ammonium, metallic (nano) composites and more [7]. Otherwise, most fouling-release coatings are constructed from silicones and fluorine-based polymers, which typically have low surface energy and low elastic modulus to make fouling organisms hardly adhere to surfaces [10]. While these coatings can significantly reduce the foulants, single antifouling strategy inevitably has more limitations: the degradation for PEG-based coatings [11], dead bacteria pollution for fouling-degrading coatings, and the poor long term stability for fouling-release coatings [12]. Hence, there is a need for the development of synergistic coatings to exhibit a combination of the above strategies [13].

Therefore, polyurethane system as a material with multi-component polymer can realize the combination of multiple antifouling strategies and has excellent mechanical properties [14]. A glycol/diamine chain extender containing specific antifouling groups is introduced into the main chain of polyurethane to achieve the antifouling effect. On the one hand, specific telechelic prepolymers were synthesized by cationic

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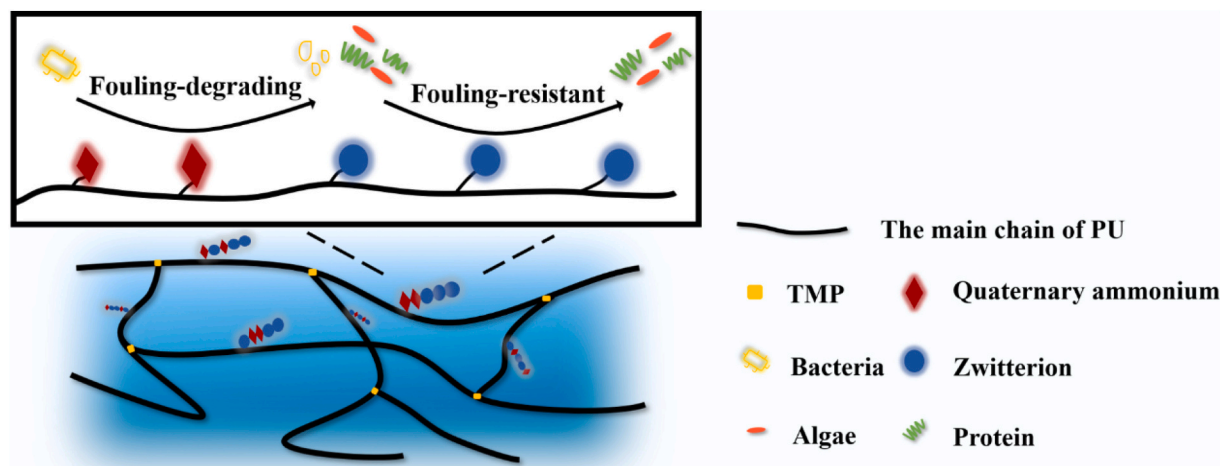


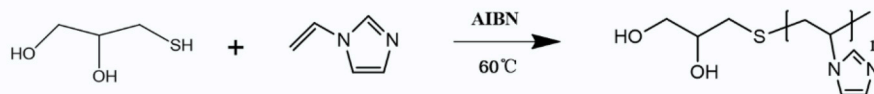
Fig. 1. Schematic illustration of the PU-PVIm-B-P coating marine antifouling mechanism.

polymerization and then were introduced into the polyurethane system [15]. In this way, the quaternary ammonium salt and PEG could be introduced to the side chain of polyurethane and the coatings exhibited high antibacterial activity with low toxicity to human cells [16]. On the other hand, the prepolymers containing specific antifouling groups with terminal dihydroxy had been successfully synthesized by thiol-ene 'click' reaction [17]. The prepolymers could contain specific functional groups like zwitterions, and could be introduced to the polyurethane system as a chain extender for the polyurethane system to perform the antifouling properties [18]. However, the chain extender containing multi-antifouling groups from both the cationic polymerization and thiol-ene 'click' reaction is hard to synthesis and improve properties. Based on these two ways, there is an urgent need to make a change to

realize concise chemical treatments and simple steps for antifouling.

In the present work, we found a way to modify the antifouling groups in the side chain from polyurethane to achieve different anti-fouling strategies and anti-fouling performances, as shown in Fig. 1. Accordingly, poly(1-Vinylimidazole) (PVIm) and its copolymers as a functional polymer [19], were easy to make nucleophilic modifications to apply in different engineering applications [20,21]. Herein, the dihydroxy-terminated poly(1-Vinylimidazole) as chain extender into polyurethane was synthesized by thiol-ene 'click' reaction. And the imidazole was modified simultaneously by 1-bromotetradecane and 1,3-propanesulfonate to obtain the quaternary ammonium and zwitterions in the side chain, which could have fouling-resistant and fouling-degrading performances underwater. Crosslinking by TMP, the

a) Synthesis of PVIm-OH



b) Synthesis of PU-PVIm-B-P

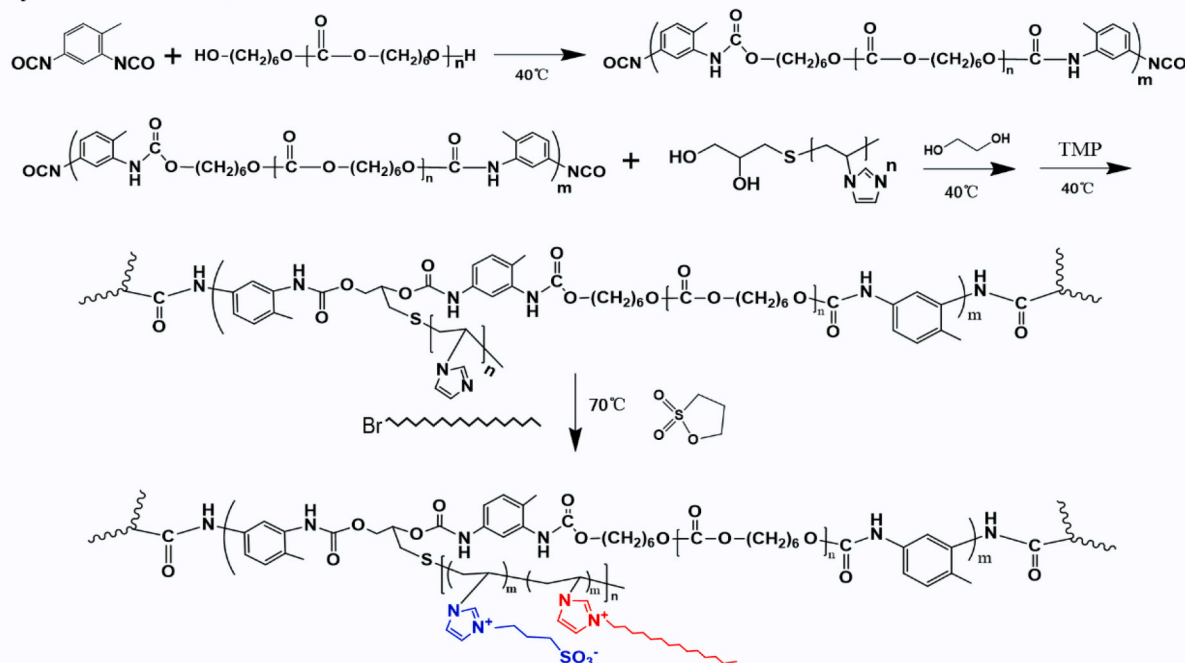


Fig. 2. Synthetic route of the polyurethane coating. (a) Synthesis of the PVIm-OH. (b) Synthesis of the PU-PVIm-B-P.

polyurethane coating has excellent comprehensive mechanical properties before swelling and low elastic modulus after swelling. The coating could produce micro/nano structure, which influenced by hydrophobic long alkyl chains from the quaternary ammonium [22] and hydrophilic zwitterions on the surface. Moreover, we could facilely adjust the amount of quaternary ammonium as well as the zwitterions to obtain excellent antifouling performances against bacteria, protein and algae.

2. Experimental

2.1. Materials

1-Thioglycerol (TG, 95 %), 2,2'-Azobis (2-methylpropionitrile) (AIBN, 98 %), 1,4-Butanediol (1,4-BD, 98 %) and 1-Vinylimidazole (VIm, 99 %) were purchased from Aladdin (Shanghai, China). Toluene Diisocyanate (TDI, 98 %) and 1,3-Propanesulfonate (PS, 99 %) were purchased from Macklin (Shanghai, China). Polycarbonate diol (PCDL, Mn = 1000) was purchased from UBE Fine Chemicals (Asia) Co. Ltd. 1-Bromotetradecane (98 %) was purchased from Innochem (Beijing, China). 1,1,1-Tris(hydroxymethyl)propane (TMP, CP) was purchased from Sinopharm Chemical Reagent Co. Ltd. Tetrahydrofuran (THF, AR), Diethyl ether (CP) and Ethyl acetate (CP) were purchased from Concord Technology (Tianjin, China). *N,N*-Dimethylformamide (DMF) was distilled from CaH₂ prior to use. All reagents were used as received.

2.2. Synthesis of dihydroxy-terminated poly(1-Vinylimidazole) (PVIIm-OH)

The dihydroxy-terminated poly (1-Vinylimidazole) (PVIIm-OH) was synthesized through the thiol-ene 'click' reaction, as shown in Fig. 2a. The TPG (0.69 g) as the chain transfer agent, VIm (3.00 g) as the monomer and AIBN (0.05%wt based on monomers) as the initiator were introduced to the system, and all of the above materials were dissolved in THF (15 mL). Vacuuming the system and filling nitrogen for three cycles, the reaction system was carried out at 70 °C for 12 h. Then cooling the system to room temperature, the solution was centrifuged and washed with ether and ethyl acetate. Finally, the product was dried under vacuum at 40 °C for 12 h.

2.3. Synthesis of polyurethane with quaternary ammonium and zwitterions (PU-PVIIm-B-P)

The PU-PVIIm-B-P was synthesized via condensation polymerization and nucleophilic reaction under nitrogen atmosphere. The DMF was used to dissolve and feed the reactants to polyurethane system. The mass ratio of polyurethane to DMF was about 1:3.

From Fig. 2b, first filling the system with nitrogen for 30 mins, then the TDI (8.98 g) and PCDL (1.70 g) were added to the system and reacted at 40 °C for 30 mins. Subsequently, the PVIIm-OH was added to the system as chain extender for polyurethane system. Upon reaction for 30 mins, the system was extended with 1,4-BD (0.31 g) for 30 mins, followed by cross-linking with TMP (14.30 mg) for 30 mins for the polymerization of polyurethane.

Then the temperature of the system was raised to 70 °C for 1 h. Following that, 1-bromotetradecane (0.38 g) and PS (0.15 g) were added to the reaction for 6 h. After the reaction was terminated, cooled the system to room temperature, washed the solution with *n*-hexane, and cured the product at 70 °C for 4 h. The feed molar ratio of the PU samples was shown in Table S1.

2.4. Characterization

2.4.1. Fourier transform infrared spectroscopy (FT-IR)

The FT-IR was performed on the Bruker VERTEX70v spectrometer at room temperature. The samples were scanned and analyzed in the wavelength range of 4000 cm⁻¹ - 400 cm⁻¹ over 64 scans.

2.4.2. Nuclear magnetic resonance hydrogen spectrum (¹H NMR)

All the samples were tested using Bucker AVIII500WB spectrometer at the temperature of 25 °C and the frequency of 500 MHz.

2.4.3. Matrix-assisted laser desorption/ionization time of flight mass spectrometry (MALDI-TOF-MS)

The accurate mass of the prepolymer was determined by Bruker Ultraflex extreme. A small amount of prepolymer sample was dissolved in methanol for analysis and test.

2.4.4. X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS, ESCALAB250XI) was used to determine the chemical composition of the samples. To calibrate the spectra, the C aliphatic signal at 284.8 eV was used.

2.4.5. Swelling ratio

The swelling ratio of polyurethane coatings after immersing in artificial seawater was analyzed by weighing rectangular samples measuring 40 × 10 × 1 mm in width and height. The swelling ratio was calculated by the formula:

$$\frac{(m_t - m_0)}{m_0} \times 100\%$$

In the formula, m_t was the mass (g) of the samples after immersing, and m_0 was the mass (g) of the samples after curing.

2.4.6. Mechanical properties and wear performance

The universal tensile testing machine was used to measure the polyurethane samples' stress-strain curve and elastic modulus. Samples were cut and fixed on the fixture, the standard distances were 20 mm and the cross-sectional areas were 10 mm². The tensile rate was 200 mm/min. The samples were tested before swelling and after swelling in artificial seawater for 7 days.

The tests of surface hardness were made according to ASTM D2240, using the LX-A Shore A scale hardness tester. The measurements were made on specimens approximately 6 mm thick. 5 measurements were made on each specimen. Herein, the samples were tested under three conditions: after curing, after swelling for 7 days in artificial seawater and drying out the coatings after swelling for 7 days [23,24].

The wear-resistance of the prepared samples was investigated by Taber wear tester. Wheel was applied with a load of 5 N pressure (250 g). The test was carried out by the rotation of capstan under sliding speed of 25 r/min. The friction wheel was H-18. The coatings were cured on the PVC substrate (5 × 5 × 0.5 cm), the weight was about 0.8 g. After being rubbed for 500 cycles, the abrasion losses of the coatings after curing and the coatings after swelling for 7 days in artificial seawater were recorded [25].

2.4.7. Contact angle and surface energy

The water contact angle (WCA) and oil angle in water (OCA) were measured by German OCA-40 Micro measuring instrument. Dropped the PU solution on a glass plate, cured it to form a film, swelled for 7 days in artificial seawater and dried it with nitrogen to remove impurities and moisture on the surface. The surface energy of the coating was calculated by the Owens-Wendt-Rabel-Kaelble method [26].

2.4.8. Surface roughness and topography

The coating surface characteristic was tested by atomic force microscopy (AFM) (Bucker Multimode8) with tapping mode at room temperature. The surface morphology and roughness of polyurethane were measured before swelling and after swelling for 7 days.

2.4.9. Antifouling performance

The antibacterial test of polyurethane samples was carried out by the spread plate method. The bacteria were gram-negative *Escherichia coli* (*E. coli*) and gram-positive *Staphylococcus aureus* (*S. aureus*). The initial

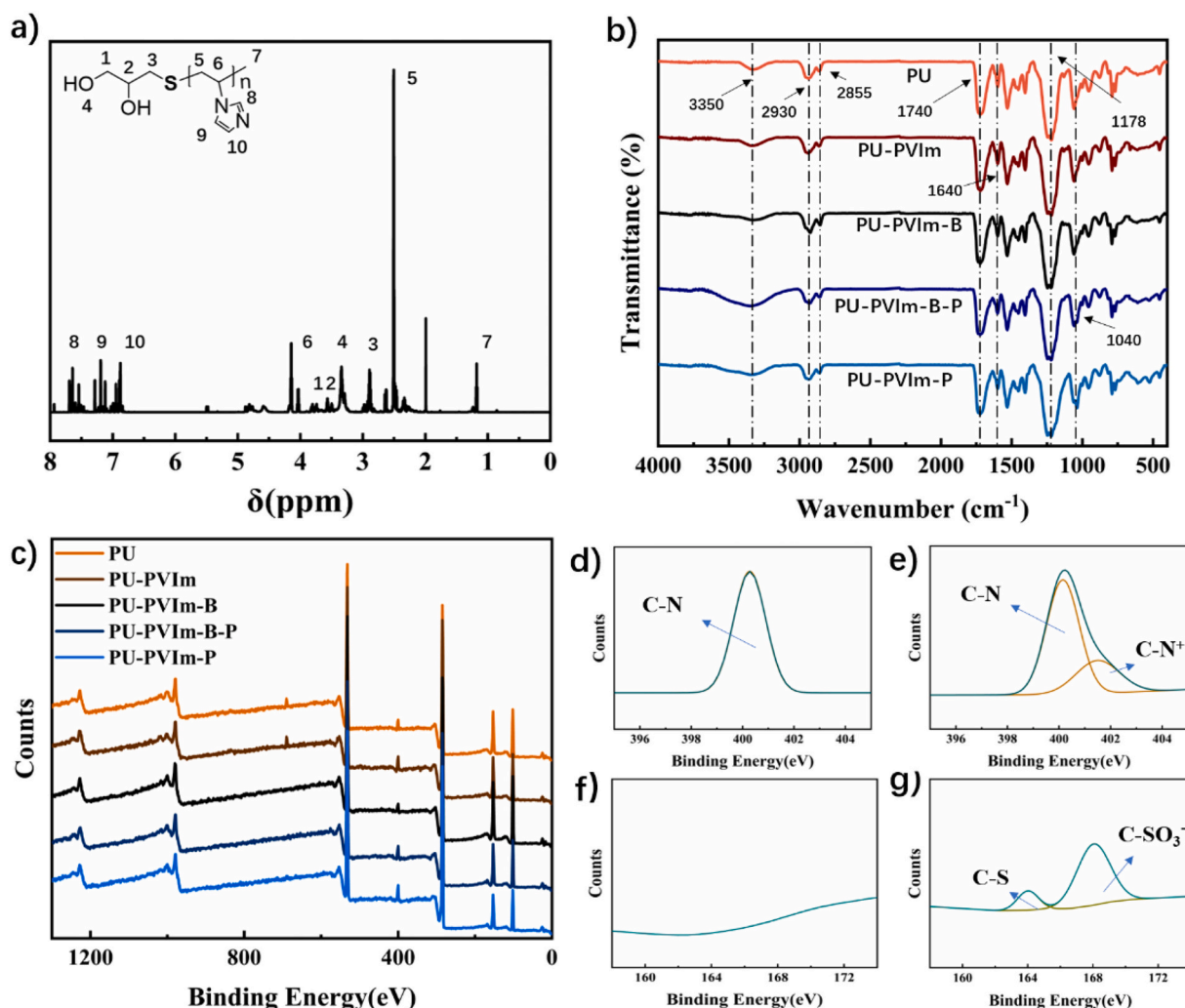


Fig. 3. The chemical structure characterization of the coatings. (a) ^1H NMR of PVIm-OH. (b) FT-IR spectra and (c) survey scan spectra of PU coatings; (d) N1s core-level spectra of PU; (e) N1s core-level spectra of PU-PVIm-B-P; (f) S2p core-level spectra of PU; (g) S2p core-level spectra of PU-PVIm-B-P.

bacterial concentration was OD 600 nm = 0.1. Before the test, all the samples were immersed in artificial seawater for 7 days and dried out with nitrogen. After diluting the coated droplets 100 times with PBS solution, the samples were incubated at 37 °C for 7 days. The number of bacteria attached to the PU coating was used as the control sample. Using the standard plate count (SPC) to analyze the bacterial adhesion rates on polyurethane coating surfaces.

The anti-protein properties of the coatings were tested with fluorescein isothiocyanate conjugated bovine serum albumin (FITC-BSA), observed by inverted fluorescence microscopy (Olympus IX83). All the samples were immersed in artificial seawater for 7 days and dried out with nitrogen, then placed in six-well plates. 5 mg/mL of FITC-BSA solution was diluted to 0.1 mg/mL with pH = 7.8 phosphate buffer, and then 3 mL of 0.1 mg/mL FITC-BSA solution were added to the six-well plates to completely cover the coatings. Incubated for 2 h at 25 °C in the dark, removed the sample, washed the surface repeatedly with phosphate buffer, moved the coatings into the fluorescence microscope to observe and took photos to record. The measuring scale in the photo was 200 μm . The area of attached BSA was counted by image J software.

The algae on the surface of the polyurethane coatings were observed and analyzed by inverted fluorescence microscopy (Olympus IX83), and the algae resistance of the coatings were evaluated with *Navicula* and *Nitzschia*. Diatoms were purchased from Xiamen Pulan Biological Technology Co., Ltd. After immersed in artificial seawater for 7 days and dried out with nitrogen, all the samples were placed in six-pore plates

with adding 10 mL diatom suspension. After incubation in a constant temperature incubator at a constant temperature of 23 °C for 72 h, every day (12 h of light and 12 h of darkness, 7 am to 7 pm for light conditions and 7 pm to 7 am for dark conditions), the surface of the polyurethane coatings was washed with artificial seawater to remove unattached diatoms. The measuring scale was 200 μm , and twenty randomly selected fields of view were counted for each sample. The number of adherent algae was counted through image J software.

Otherwise, herein we just selected the PU, PU-PVIm-B-P and PU-PVIm-P coatings to investigate the antifouling performances in the actual marine environment. The long-term antifouling test in the marine environment was tested in Da Lian (China) for 3 months (from May to Aug). Three samples were evenly sprayed on the PVC substrate (10 * 15 * 0.5 cm, actual area was 9*14 cm). All samples were fixed in one iron shelf, and the depth was about 0.2–1 m, the samples were facing the sunlight. After the test, we took photos to observe the coatings surfaces.

3. Discussion

3.1. Chemical characterization of the prepolymer PVIm-OH

As shown in Fig. S2, FT-IR spectra were recorded for TG, VIm and PVIm-OH. The absorption bands at 729 cm^{-1} were assigned to the vibration of the $-\text{CH}_2-$ groups in the long chain from VIm, and the absorption peaks at 3450 cm^{-1} were assigned to the vibration of the $-\text{OH}$

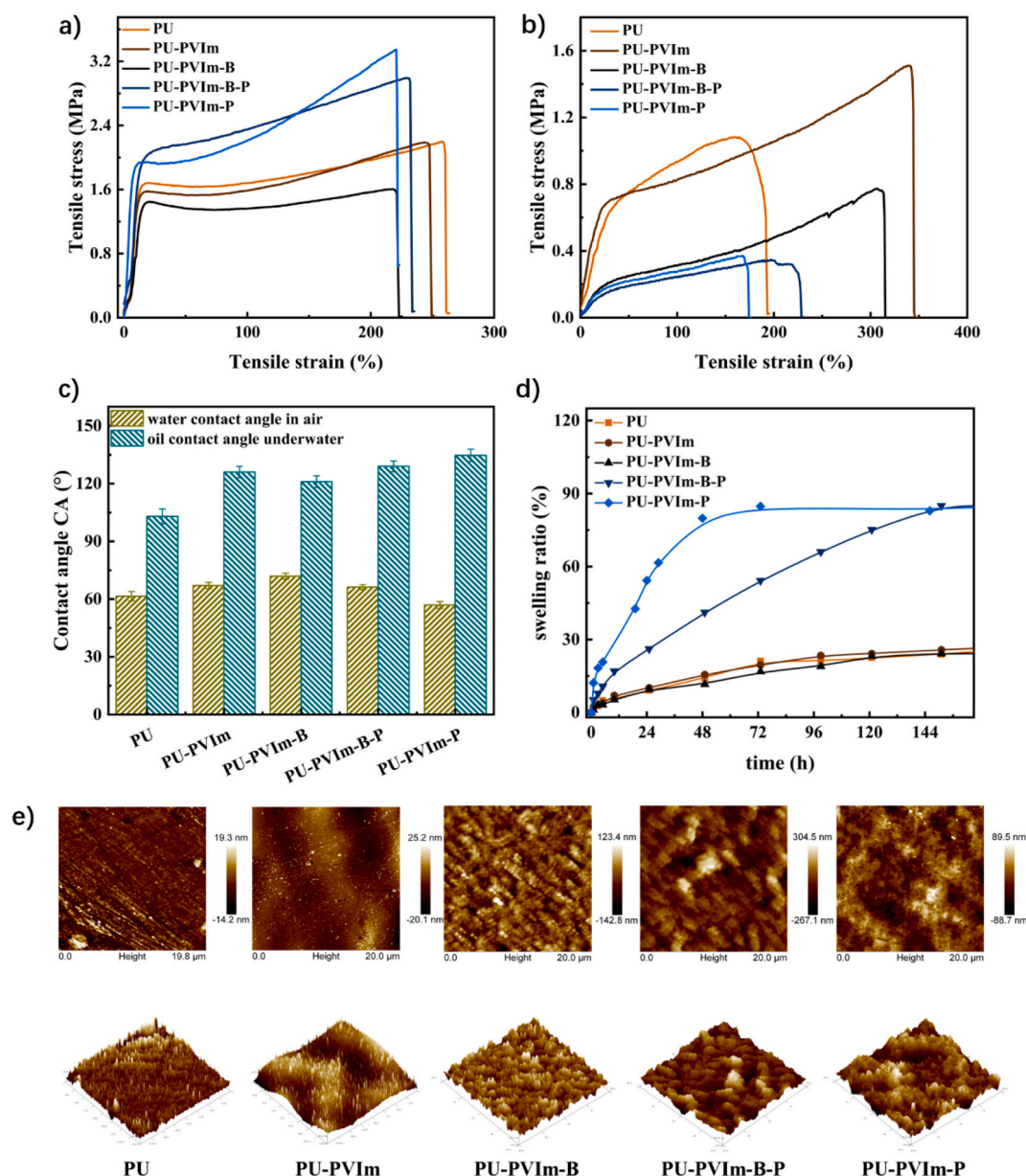


Fig. 4. The mechanical properties, swelling ratio and surface performances of the PU coatings. The stress and strain curves of the PU coatings (a) after curing and (b) after swelling for 7 days. (c) the swelling rate of the PU coatings. (d) the water contact angle and the oil contact angle underwater of PU coatings. (e) the AFM of PU coatings after swelling for 7 days.

groups from TG. The characteristic peaks at about 3110 cm^{-1} were assigned to the stretching vibration of the $=\text{C-H}$ from the imidazole, and the bands observed at 1647 cm^{-1} were attributed to $\text{C}=\text{C}$ from vinylimidazole [22].

Fig. 3a exhibited the ^1H NMR spectrum of PVIm-OH. The peaks at 1.1 ppm represented the $-\text{CH}_3$ at the end of the alkyl backbone, and two signals at 2.55 ppm and 4.17 ppm belonged to the $-\text{CH}$ - and $-\text{CH}_2$ - on the prepolymer formed by vinylimidazole. In addition, the characteristic peaks at 6.97 ppm, 7.21 ppm and 7.62 ppm coming from the imidazole ring appeared. The peaks at 2.89 were the proton characteristic peaks of the $-\text{CH}_2$ - groups on the main chain. And the peaks at 3.55 ppm and 3.71 ppm were ascribed to the $-\text{CH}$ - and $-\text{CH}_2$ - groups connected with the hydroxyl group, respectively. Moreover, the characteristic peak of $-\text{OH}$ was at 3.28 ppm. Furthermore, the signals at 5–6 ppm corresponding to the two protons of the vinyl group disappeared, which confirms that the prepolymer PVIm-OH was successfully synthesized [22,27].

Otherwise, the mass of the prepolymer was measured through

MALDI-TOF-MS. According to the results (Fig. S3), the repeat unit ($\text{RU} = 94$) could be well-identified. The analysis of the PVIm-OH main series provided the information on the structure. The results indicated that the prepolymer was mainly formed by the polymerization of one TG group with four vinylimidazole monomers for peaks A and one TG group with five vinylimidazole monomers for peaks B. These results indicated that the PVIm-OH was successfully synthesized [28].

3.2. Chemical characterization of the polyurethane coating PU-PVIm-B-P

FT-IR spectra (Fig. 3b) were recorded for the PU coatings. The characteristic peaks at 3350 cm^{-1} that referred to the impurity water appeared on all samples, and the peaks at 2930 cm^{-1} and 2855 cm^{-1} were mainly from the stretching vibration of the asymmetric and symmetric methyl. Furthermore, the adsorption bands at 1740 cm^{-1} were due to the stretching vibration of the $\text{C}=\text{O}$ from the main chain of polyurethane. At 1640 cm^{-1} , the bands were the stretching vibration of

the C=C from the benzene in TDI which was the monomer for synthesizing polyurethane. Additionally, at 1178 cm^{-1} , the peaks were associated with the stretched vibrations of the tertiary amines [22]. And at 1040 cm^{-1} , the peaks corresponded to the sulfonic group which was generated by the reaction between the imidazole and PS [29].

The ^1H NMR (Fig. S4) was recorded for PU-PVIm-B-P. It was shown that PCDL was successfully included in the polyurethane reaction as a soft segment with the characteristic peaks at 4.21 ppm, 1.62 ppm and 1.43 ppm, corresponding to the $-\text{CH}_2-$ under different chemical shifts. And at 2.21 ppm, the signals were from the proton peaks of the benzene, meaning that TDI was successfully involved in the reaction. The characteristic peaks at 3.86 ppm represented the $-\text{CH}_2-$ from the chain extender formed by vinylimidazole. Moreover, at 1.29 ppm, the peaks were ascribed to $-\text{CH}_2-$ on the long alkyl chain formed by 1-bromotetradecane as well as the peaks at 0.88 ppm were attributed to $-\text{CH}_3$. These results proved that the PVIm-OH had successfully participated in the reaction of the polyurethane system, and the modification of imidazole has been realized successfully [30,31].

3.3. X-ray photoelectron spectroscopic analysis

The surface composition analysis of PU coatings was determined by XPS from Fig. 3c. There were two main peaks in the wide scan XPS spectrum that corresponded to C1s (285 eV) and O1s (531 eV). Two peaks could be decomposed from the N1s (400 eV) and the S2p (166 eV) [31].

As shown in Fig. 3d, the N1s peaks at 399 eV were attributed to the =N-H- groups in PU, while the PU-PVIm-B-P peaks at 399 eV and 402 eV from Fig. 3e were associated with the =N-H- groups from PU and quaternary ammonium groups from the modification by 1-bromotetradecane, respectively [32]. Similarly, by comparing Fig. 3f and g, the S2p could be separated into two peaks: the sulfonic acid group at 169.5 eV from zwitterions and the C-S at 164.9 eV from the prepolymer which synthesized by TG [33]. These results indicated that the quaternary ammonium and zwitterions have been successfully modified via nucleophilic reaction.

3.4. Mechanical and wear properties of the coatings

The tensile tests were used to determine the mechanical properties of the PU coatings before and after swelling for 7 days. As shown in Figs. 4a and S5, after curing, the tensile strength and elastic modulus of PU were 2.20 MPa and 9.75 MPa respectively. PU-PVIm has the same mechanical properties as PU after the introduction of PVIm-OH. Additionally, the decrease in tensile strength and elastic modulus after the imidazole modified by 1-bromotetradecane confirmed that the mechanical properties of straight-chain polymers was generally superior to those of polymers with side chain [34]. It was fascinating to note that PU-PVIm-P has the highest tensile strength (3.34 MPa) and elastic modulus (14.69 MPa), followed by PU-PVIm-B-P. These results confirmed that the zwitterions in side chains could form strong ionic bonds between the polymer segments which increased the mechanical properties [35].

Otherwise, from Figs. 4b and S5, the tensile strength and elastic modulus of PU decreased to 2.20 MPa and 9.75 MPa after swelling for 7 days. The PU-PVIm showed a significantly increase in tensile strength (1.50 MPa) and elastic modulus (2.03 MPa), which might be explained by the fact that the water molecules could easily form hydrogen bonds between the imidazole in the side chain [36,37]. Similarly, the mechanical properties of PU-PVIm-B decreased due to the hydrophobic side chain and swelling. As the result of the ionic bonds dissolved in water, the tensile strength and elastic modulus were reduced in PU-PVIm-B-P and PU-PVIm-P [38]. However, the low modulus was conducive to the removal of marine microorganisms on the surface.

From the Fig. S6, we could see that the hardness of the coatings could be around 55 HA after curing, like rubber [39]. And the hardness of coatings after swelling for 7 days was around 37 HA. Then we dried out

the coatings after swelling for 7 days, the hardness could return to around 45 HA, which meant that the hardness could be restored to around 82 % compared to the coatings after curing [23,24].

Otherwise, from Fig. S7, the abrasion losses of the coatings after curing were 0.01, 0.04, 0.05, 0.03 and 0.04 g respectively. This result proved that the coatings had good wear-resistance. After swelling for 7 days, the abrasion losses of the coatings were 0.04, 0.04, 0.12, 0.18 and 0.23 g. Even though the swelling of the hydrophilic coatings led to the increase of abrasion loss, the coating could retain around 70 % of the components [25].

3.5. Surface wettability of the coatings

The water dynamic contact angles (WCA), the oil contact angles underwater (OCA) and the surface energy of the coatings were used to evaluate the wettability mechanism. According to the results (Fig. 4c and S8), the WCA for PU, PU-PVIm and PU-PVIm-B were 61.5° , 67.0° and 71.9° respectively. Moreover, with the introduction of PVIm and hydrophobic 1-bromotetradecane to the side chain, the results concluded that the coatings exhibited hydrophobicity and lipophilicity (the OCA was 126° , 111.4° and 126° respectively). Hence, the surface energy decreased from 41.5 mN/m for PU to 34.2 mN/m for PU-PVIm-B. With the percentage of zwitterions increasing after the modification, the WCA advanced and the OCA reduced in PU-PVIm-B-P and PU-PVIm-P. Similarly, the PU-PVIm-B-P surface energy increased as well as the PU-PVIm-P. Through Wenzel relation the Owens-Wendt-Rabel-Kaelble method, we could find that the surface energy has an obvious change in PU-PVIm-P [40–42]. The results indicated that hydrophobic quaternary ammonium in this system could increase hydrophilicity and had lower surface energy, while the zwitterions had an effect on hydrophilicity and possessed higher surface energy for coatings [8].

3.6. Swelling ratio of the coatings

Fig. 4e illustrated the variation in the swelling ratio of the coatings over time. All of the PU coatings were immersed in artificial seawater for 7 days at room temperature. As observed from the figure, the swelling ratio of PU, PU-PVIm and PU-PVIm-B were 25 %, 27 % and 24 % after reaching the swelling equilibrium. These results confirmed that the introduction of PVIm and hydrophobic 1-bromotetradecane on the side chain had little effect on the hydrophilicity of PU, as well as the swelling ratio. On the contrary, in the case of PU-PVIm-B-P and PU-PVIm-P, the swelling ratio seemed to significantly increase to 85 % and 85 %, which might be due to the modification of zwitterions. In addition, the swelling speed of PU-PVIm-B-P was relatively slower than PU-PVIm-P due to the restraint of the hydrophobic unit.

3.7. Surface roughness and topography

Under marine environments, the changes in surface roughness and topography were primarily caused by the modifications of hydrophilic and hydrophobic units in the side chain, which impacted the release of fouling organisms [43,44]. Fig. S9, Figs. 4e and S10 showed the 2D, 3D figures and the corresponding Ra values for the PU coatings before and after swelling for 7 days. The PU, PU-PVIm, PU-PVIm-B, PU-PVIm-B-P and PU-PVIm-P surfaces showed the Ra values of 0.61 nm, 0.81 nm, 0.50 nm, 0.62 nm and 1.06 nm respectively before swelling. In these coatings, the surface roughness remained relatively stable, which dedicated that the modifications on the side chain hardly affect the surface morphology after curing.

Otherwise, the surface roughness of the coatings was found to have different performances under wet condition. Due to the swelling of the coating after being immersed in artificial seawater for 7 days, the Ra value for PU increased to 2.9 nm. Subsequently, the VIm and 1-bromotetradecane were introduced into the side chain, the Ra value increased to 4.6 nm and 28.6 nm respectively. Due to the polarity difference

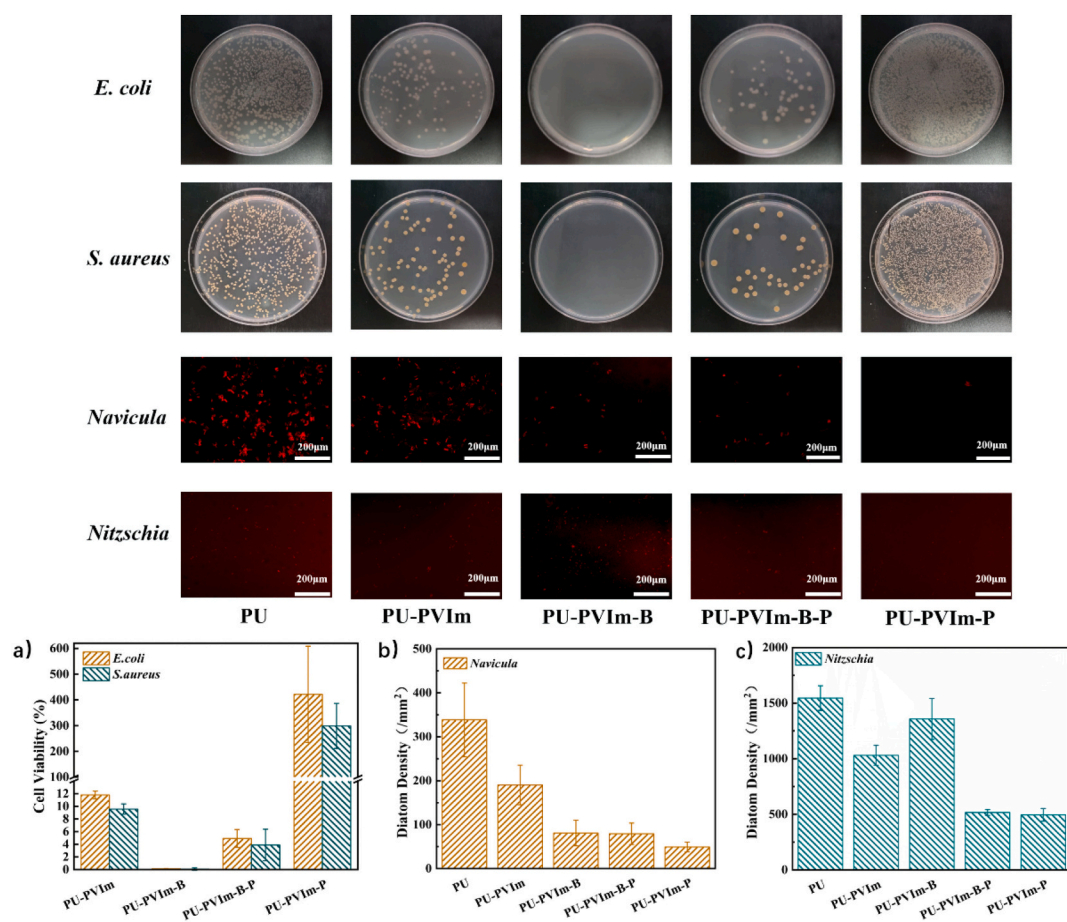


Fig. 5. The marine antifouling properties of different PU coatings and substrates. (a) the cell viability of *E. coli* and *S. aureus*. The Diatom Density of (b) *Navicula* and (c) *Nitzschia*.

between the hydrophobic quaternary ammonium formed in the side chain and the hydrophilic polyurethane main chain, there was a slight phase separation on the surface of PU-PVIm-B [22]. Similarly, the maximum roughness (41.5 nm) of PU-PVIm-B-P could be explained by a combination of two parts: the modification of hydrophobic quaternary ammonium and the obvious swelling caused by zwitterions. Consequently, the results led to a larger phase separation occurring on the coating surface than PU-PVIm-B. With regard to the PU-PVIm-P, the *Ra* value was 17.1 nm, which might be mainly affected by the swelling.

This study confirmed that the modification by hydrophobic quaternary ammonium and hydrophilic zwitterions in the side chain could lead a slight phase separation on the surface, exhibited micro/nano structure, which might be beneficial to marine anti-fouling [45].

3.8. Antifouling properties of the coatings

Bacteria were among the series of marine fouling organisms. The coatings PU-PVIm, PU-PVIm-B, PU-PVIm-B-P and PU-PVIm-P were evaluated for their antibacterial activities against *E. coli* and *S. aureus*. The number of bacteria adhering to the PU coating surface was used as the control sample. According to the results (Fig. 5a), PU-PVIm showed antibacterial activity of approximately 12 % and 11 % against *E. coli* and *S. aureus* respectively, indicating that imidazole could perform excellent antibacterial effects. For PU-PVIm-B, the cell viability of both *E. coli* and *S. aureus* remarkably decreased to basically 0 % with the modification from imidazole to quaternary ammonium in the side chain. Superior antibacterial properties were demonstrated by the effect of quaternary ammonium on bacteria [46]. In the same manner, PU-PVIm-B-P was found to have excellent antibacterial properties due to the quaternary

ammonium, which exhibited 4 % and 5 % cell viability against *E. coli* and *S. aureus* respectively. However, the bacteria were more likely to adhere to the PU-PVIm-P compared with PU due to the hydrophilicity and high surface energy after introducing zwitterions [47].

PU coatings were also evaluated for their anti-protein properties against bovine serum albumin (BSA). Similarly, the area of BSA adhering to the PU coating surface was used as the control sample. From Fig. S11, the protein adhesive area of PU-PVIm reduced to 52 % compared to PU. And the adhesion percentage of protein for PU-PVIm-B and PU-PVIm-B-P is 17 % and 16 % respectively, which was related to the low surface energy and the hydration layer from zwitterions. The protein adhesion was 41 % for the PU-PVIm-P, which was attributed to zwitterionic groups used for resisting the protein [48]. Based on the results, it was observed that the surface energy [49] and fouling resistance properties of zwitterions had a great impact on anti-protein characteristics.

The attachment of algae to the surface could also lead to surface damage in the marine fouling. In this work, the *Navicula* and *Nitzschia* were used to test the anti-algae performance of the PU coatings. From Fig. 5b and c, after the introduction of imidazole into the side chain, the antibacterial properties of imidazole and the low surface area of the coating reduced a little number of *Navicula* and *Nitzschia*. After the imidazole ring was modified by 1-bromotetradecane, the low surface of the PU-PVIm-B coating could also improve its anti-algae performance. As the multifunctional anti-fouling coating PU-PVIm-B-P, the properties of anti-fouling release of zwitterions were increased on the base of the quaternary ammonium. The adhesion of algae decreased obviously. As for PU-PVIm-P, the polyurethane coating with only zwitterions in the side chain could also perform excellent anti-algae characteristics. The result indicated the key role of zwitterions for anti-algae [50].

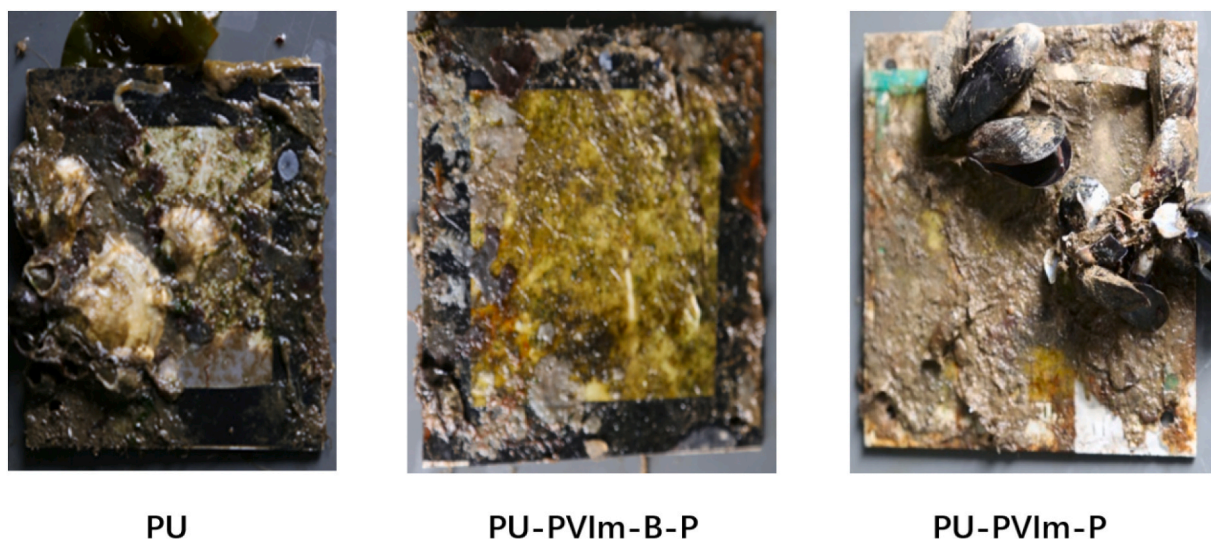


Fig. 6. Long-term actual marine antifouling experiment: the photographs of the actual marine anti-biofouling performance (no cleaning) from PU, PU-PVIm-B-P and PU-PVIm-P.

Otherwise, we took long-term marine anti biofouling test to observe the marine antifouling performance. The results showed the long-term actual photographs of ocean pollution (no cleaning). As showed in Fig. 6, we could see that the number of barnacles reduced in the coatings surface after modification, but the barnacles still could adhesive to the coatings. The microfouling film could form in every coating surface, meanwhile the macrofouling such as barnacles and mussels had already grown on the coating surfaces [51]. Therefore, there were still some limitations for the effect of the coatings in the actual marine environment.

Based on the above experiments for anti-bacterial, anti-protein and anti-algae, the treated coatings exhibited excellent comprehensive antifouling effects. On the one hand, the quaternary ammonium from side chain performed excellent fouling-degrade property against bacteria. On the other hand, the algae and protein were resistant because of the hydrated layer from zwitterionic groups. However, the treated coatings surface still had large number of marine organisms adhering on the coated surface. The coatings effect still had some limitations in the actual marine environment.

4. Conclusions

In this work, we synthesized a novel antifouling polyurethane coating with fouling degrading group quaternary ammonium and fouling-resistant group zwitterions as the side chain. The quaternary ammonium and zwitterions were obtained by modification on the imidazole from the side chain of polyurethane, where the PVIm-OH was synthesized by thiol-ene 'click' reaction and took part in the polyurethane system as specific chain extender. The multiple antifouling strategies could be obtained by simple modification rather than direct combination. Because the quaternary ammonium was relatively hydrophobic and zwitterions were relatively hydrophilic, the coating could show regional micro/nano structures. Otherwise, quaternary ammonium and zwitterions played a significant role in fouling-degrading (against Bacteria) and fouling-resistant (against BSA and algae) properties respectively. However, due to the complexity of the marine environment, there was still a long road to make this coating into marine industry.

Availability of data and material

All data in this study are included in this published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2022.107225>.

References

- [1] M.P. Schultz, J.A. Bendick, E.R. Holm, W.M. Hertel, Economic impact of biofouling on a naval surface ship, *Biofouling* 27 (2011) 87–98.
- [2] C.M. Duarte, S. Agusti, E. Barbier, G.L. Britten, J.C. Castilla, J.P. Gattuso, R. W. Fulweiler, T.P. Hughes, N. Knowlton, C.E. Lovelock, H.K. Lotze, M. Predragovic, E. Poloczanska, C. Roberts, B. Worm, Rebuilding marine life, *Nature* 580 (2020) 39–51.
- [3] C.M. Duarte, M. Holmer, Y. Olsen, D. Soto, N. Marbà, J. Guieu, K. Black, I. Karakassis, Will the oceans help feed humanity? *Bioscience* 59 (2009) 967–976.
- [4] C.M. Roberts, B.C. O'Leary, D.J. McCauley, P.M. Cury, C.M. Duarte, J. Lubchenco, D. Pauly, A. Sáenz-Arroyo, U.R. Sumaila, R.W. Wilson, Marine reserves can mitigate and promote adaptation to climate change, *Proc. Natl. Acad. Sci.* 114 (2017) 6167–6175.
- [5] J.-P. Gattuso, A.K. Magnan, L. Bopp, W.W. Cheung, C.M. Duarte, J. Hinkel, E. Mcleod, F. Micheli, A. Oschlies, P. Williamson, Ocean solutions to address climate change and its effects on marine ecosystems, *Front. Mar. Sci.* 5 (2018) 337.
- [6] V. Eyring, H.W. Köhler, J. Van Aardenne, A. Lauer, Emissions from international shipping: 1. The last 50 years, *J. Geophys. Res. D: Atmos.* 110 (2005) 171–182.
- [7] A.M.C. Maan, A.H. Hofman, W.M. Vos, M. Kamperman, Recent developments and practical feasibility of polymer-based antifouling coatings, *Adv. Funct. Mater.* 30 (2020) 2000936.
- [8] H. Qiu, K. Feng, A. Gapeeva, K. Meurisch, S. Kaps, X. Li, L. Yu, Y.K. Mishra, R. Adelung, M. Baum, Functional polymer materials for modern marine biofouling control, *Prog. Polym. Sci.* 127 (2022), 101516.
- [9] S. Pradhan, S. Kumar, S. Mohanty, S.K. Nayak, Environmentally benign fouling-resistant marine coatings: a review, *Polym. Plastics Technol. Mater.* 58 (2018) 498–518.

- [10] A.K. Leonardi, C.K. Ober, Polymer-based marine antifouling and fouling release surfaces: strategies for synthesis and modification, *Annual review of chemical and biomolecular engineering* 10 (2019) 241–264.
- [11] L. Zheng, H.S. Sundaram, Z. Wei, C. Li, Z. Yuan, Applications of zwitterionic polymers, *React. Funct. Polym.* 118 (2017) 51–61.
- [12] G.P. Sakala, M. Reches, Peptide-based approaches to fight biofouling, *Adv. Mater. Interfaces* 5 (2018) 1800073.
- [13] P. Shivapooja, Q. Yu, B. Orihuela, R. Mays, D. Rittschof, J. Genzer, G.P. López, Modification of silicone elastomer surfaces with zwitterionic polymers: short-term fouling resistance and triggered biofouling release, *ACS Appl. Mater. Interfaces* 7 (2015) 25586–25591.
- [14] A. Atiqah, M.T. Mastura, B.A.A. Ali, M. Jawaid, S.M. Sapuan, A review on polyurethane and its polymer composites, *Curr. Org. Synth.* 14 (2017) 233–248.
- [15] T. Fujiwara, U. Makal, K.J. Wynne, Synthesis and characterization of novel amphiphilic telechelic polyoxetanes, *Macromolecules* 36 (2003) 9383–9389.
- [16] C. Wang, O. Zolotarevskaya, K.M. Ashraf, X. Wen, D.E. Ohman, K.J. Wynne, Surface characterization, antimicrobial effectiveness, and human cell response for a biomedical grade polyurethane blended with a mixed soft block PTMO-Quat/PEG copolyoxetane polyurethane, *ACS Appl. Mater. Interfaces* 11 (2019) 20699–20714.
- [17] C. Ma, H. Zhou, B. Wu, G. Zhang, Preparation of polyurethane with zwitterionic side chains and their protein resistance, *ACS Appl. Mater. Interfaces* 3 (2011) 455–461.
- [18] J. Wu, C. Wang, Y. Xiao, C. Mu, W. Lin, Fabrication of water-resistance and durable antimicrobial adhesion polyurethane coating containing weakly amphiphilic poly (isobornyl acrylate) side chains, *Prog. Org. Coat.* 147 (2020), 105812.
- [19] H. Gorjian, H. Fahim, N.G. Khaligh, Poly (N-vinylimidazole): a biocompatible and biodegradable functional polymer, metal-free, and highly recyclable heterogeneous catalyst for the mechanochemical synthesis of oximes, *Turk. J. Chem.* 45 (2021) 2007–2012.
- [20] A. Basaiahgari, S.K. Yadav, R.L. Gardas, Zwitterions as novel phase forming components of aqueous biphasic systems, *Pure Appl. Chem.* 91 (2019) 1279–1294.
- [21] L.A. Camacho-Cruz, M.A. Velasco-Medel, H. Parra-Delgado, E. Bucio, Functionalization of cotton gauzes with poly(N-vinylimidazole) and quaternized poly(N-vinylimidazole) with gamma radiation to produce medical devices with pH-buffering and antimicrobial properties, *Cellulose* 28 (2021) 3279–3294.
- [22] D. Hao, X. Wang, X. Liu, X. Zhu, S. Sun, J. Li, O. Yue, A novel eco-friendly imidazole ionic liquids based amphoteric polymers for high performance fatliquoring in chromium-free tanned leather production, *J. Hazard. Mater.* 399 (2020), 123048.
- [23] A. Siddiqui, M. Braden, M.P. Patel, S. Parker, An experimental and theoretical study of the effect of sample thickness on the shore hardness of elastomers, *Dent. Mater.* 26 (2010) 560–564.
- [24] R. Jia, X. Liu, Z. Huang, D. Wang, C. Zhao, Z. Hui, X. He, D. Wu, Novel polyurethane elastomer modified by hybrid Shell Nano-/Microcapsules for unique self-lubricating behavior, *Ind. Eng. Chem. Res.* 59 (2020) 22123–22131.
- [25] Z. Liu, H. Wang, X. Zhang, C. Lv, C. Wang, Y. Zhu, Robust and chemically stable superhydrophobic composite ceramic coating repellent even to hot water, *Adv. Mater. Interfaces* 4 (2017) 1601202.
- [26] D.K. Owens, R. Wendt, Estimation of the surface free energy of polymers, *J. Appl. Polym. Sci.* 13 (1969) 1741–1747.
- [27] Q. Xie, H. Zeng, Q. Peng, C. Bressy, C. Ma, G. Zhang, Self-stratifying silicone coating with nonleaching antifoulant for marine anti-biofouling, *Adv. Mater. Interfaces* 6 (2019) 1900535.
- [28] X. Yang, T. Wu, B. Liu, Y. Du, H. Li, S. Zhao, Y. Lu, Matrix selection for polymer guanidine analysis by MALDI-TOF MS, *Int. J. Mass Spectrom.* 356 (2013) 1–6.
- [29] D. Lin-Vien, N.B. Colthup, W.G. Fateley, J.G. Grasselli, *The handbook of infrared and raman characteristic frequencies of organic molecules*, Elsevier, 1991.
- [30] Q. Xie, C. Ma, C. Liu, J. Ma, G. Zhang, Poly(dimethylsiloxane)-based polyurethane with chemically attached antifoulants for durable marine antibiofouling, *ACS Appl. Mater. Interfaces* 7 (2015) 21030–21037.
- [31] Y. Wang, J. Du, H. Guo, R. Liu, Z. Li, T. Yang, J. Ai, C. Liu, The antibacterial activity and mechanism of polyurethane coating with quaternary ammonium salt, *J. Polym. Res.* 29 (2022) 1–13.
- [32] W. Cao, Z. Wang, Q. Zeng, C. Shen, 13C NMR and XPS characterization of anion adsorbent with quaternary ammonium groups prepared from rice straw, corn stalk and sugarcane bagasse, *Appl. Surf. Sci.* 389 (2016) 404–410.
- [33] J. Huang, W. Xu, Zwitterionic monomer graft copolymerization onto polyurethane surface through a PEG spacer, *Appl. Surf. Sci.* 256 (2010) 3921–3927.
- [34] H. Sheikhy, M. Shahidzadeh, B. Ramezanzadeh, F. Noroozi, Studying the effects of chain extenders chemical structures on the adhesion and mechanical properties of a polyurethane adhesive, *J. Ind. Eng. Chem.* 19 (2013) 1949–1955.
- [35] X. Li, C. Tang, D. Liu, Z. Yuan, H.C. Hung, S. Luozhong, W. Gu, K. Wu, S. Jiang, High-strength and nonfouling zwitterionic triple-network hydrogel in saline environments, *Adv. Mater.* 33 (2021), e2102479.
- [36] J. Smets, W. McCarthy, G. Maes, L. Adamowicz, Correlations between ab initio and experimental data for isolated 1:1 hydrogen-bonded complexes of pyridine and imidazole derivatives with water, *J. Mol. Struct.* 476 (1999) 27–43.
- [37] X.N. Zhang, Y.J. Wang, S. Sun, L. Hou, P. Wu, Z.L. Wu, Q. Zheng, A tough and stiff hydrogel with tunable water content and mechanical properties based on the synergistic effect of hydrogen bonding and hydrophobic interaction, *Macromolecules* 51 (2018) 8136–8146.
- [38] H. Wang, Y. Hu, D. Lynch, M. Young, S. Li, H. Cong, F.J. Xu, G. Cheng, Zwitterionic polyurethanes with tunable surface and bulk properties, *ACS Appl. Mater. Interfaces* 10 (2018) 37609–37617.
- [39] S. Fang, Y. Hu, L. Song, J. Zhan, Q. He, Mechanical properties, fire performance and thermal stability of magnesium hydroxide sulfate hydrate whiskers flame retardant silicone rubber, *J. Mater. Sci.* 43 (2007) 1057–1062.
- [40] R.N. Wenzel, Resistance of solid surfaces to wetting by water, *Ind. Eng. Chem.* 28 (1936) 988–994.
- [41] D. Quéré, Wetting and roughness, *Annu. Rev. Mater. Res.* 38 (2008) 71–99.
- [42] K.J. Varin, J. Cohen, Wettability of terminally anchored polymer brush layers on a polyamide surface, *J. Colloid Interface Sci.* 436 (2014) 286–295.
- [43] L. Xiao, J.A. Finlay, M. Röhrig, S. Mieszkun, M. Worgull, H. Hölscher, J.A. Callow, M.E. Callow, M. Grunze, A. Rosenhahn, Topographic cues guide the attachment of diatom cells and algal zoospores, *Biofouling* 34 (2018) 86–97.
- [44] L. Xiao, S.E.M. Thompson, M. Röhrig, M.E. Callow, J.A. Callow, M. Grunze, A. Rosenhahn, Hot embossed microtopographic gradients reveal morphological cues that guide the settlement of zoospores, *Langmuir* 29 (2013) 1093–1099.
- [45] Z.-P. Zhang, X.-F. Song, L.-Y. Cui, Y.-H. Qi, Synthesis of polydimethylsiloxane-modified polyurethane and the structure and properties of its antifouling coatings, *Coatings* 8 (2018) 157.
- [46] C. Fang, L. Kong, Q. Ge, W. Zhang, X. Zhou, L. Zhang, X. Wang, Antibacterial activities of N-alkyl imidazolium-based poly(ionic liquid) nanoparticles, *Polym. Chem.* 10 (2019) 209–218.
- [47] S.Y. Zheng, S. Mao, J. Yuan, S. Wang, X. He, X. Zhang, C. Du, D. Zhang, Z.L. Wu, J. Yang, Molecularly engineered zwitterionic hydrogels with high toughness and self-healing capacity for soft electronics applications, *Chem. Mater.* 33 (2021) 8418–8429.
- [48] S. Davari, M. Omidkhah, M. Abdollahi, Improved antifouling ability of thin film composite polyamide membrane modified by a pH-sensitive imidazole-based zwitterionic polyelectrolyte, *J. Membr. Sci.* 564 (2018) 788–799.
- [49] R.E. Baier, Surface behaviour of biomaterials: the theta surface for biocompatibility, *J. Mater. Sci. Mater. Med.* 17 (2006) 1057–1062.
- [50] X. Su, D. Hao, X. Xu, X. Guo, Z. Li, L. Jiang, Hydrophilic/Hydrophobic heterogeneity anti-biofouling hydrogels with well-regulated rehydration, *ACS Appl. Mater. Interfaces* 12 (2020) 25316–25323.
- [51] X. Su, M. Yang, D. Hao, X. Guo, L. Jiang, Marine antifouling coatings with surface topographies triggered by phase segregation, *J. Colloid Interface Sci.* 598 (2021) 104–112.